

Marena Abboud

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EDUCATION

Moravian University, Bethlehem, PA

May 2026

B.S. Computer Science, Minor in Art · GPA 3.88 · Summa Cum Laude · Dean's List 2022–2026

PROJECT EXPERIENCE

MindScape 360 — VR Mindfulness App | github.com/MoravianUniversity/Mindscape-360 — SOAR Fellowship, Moravian University Summer 2025

- Designed and published a cross-platform VR mindfulness app for acute-care patients, built to minimize onboarding steps so users in pain can start sessions right away.
- Partnered with a Doctor of Physical Therapy to align the UI with clinical research on accessibility and stress reduction. Built in Kotlin and Jetpack Compose with C++, integrating MediaPlayer on Android and AVPlayer on iOS for synchronized guided audio and visuals.
- Published on the App Store and Google Play and currently piloting at Good Shepherd Rehabilitation Hospital. Named a URSCA All-Star by Moravian's Rehabilitation Sciences Occupational Therapy program. Also built companion apps including Tic-Tac-Toe, a Misophonia survey tool, and a Sketchpad app for Android and iOS.

Mirrulations Search — Federal Regulations Search Tool | github.com/mirrulations/mirrulations-search — Client-Sponsored Capstone, Moravian University Fall 2025 – Spring 2026

- Worked directly with clients from CareSet and Merck in a real-world agile environment to build a search platform that helps researchers and policy professionals navigate federal rulemaking by searching public dockets, documents, and comments from regulations.gov. Collaborated across specialized teams using a shared GitHub workflow with code reviews, pull requests, and sprint-based releases.
- Designed and implemented the full document download system from scratch, including REST API endpoints, a Redis-backed async worker for packaging and uploading files to S3, a background job for cleanup, PostgreSQL schema changes, EC2 deployment with systemd, and frontend integration.
- Also implemented Google OAuth for user authentication, CFR part filtering for targeted regulatory search, and sort-by functionality across date, comment count, and document count. Tech stack: Python, Flask, PostgreSQL, OpenSearch, Redis, AWS S3/EC2, React.

BytePath — Full-Stack Learning Platform — Moravian University Fall 2025 – Present

- Built a production Python learning platform used by ~30 students across two intro CS courses at Moravian, featuring leveled question sets, per-student progress tracking, and instructor dashboards.
- Built a Flask and SQLAlchemy REST API with 6 route blueprints backed by SQLite, paired with a React and TypeScript frontend. Integrated Google OAuth and deployed to AWS EC2 with automated systemd services.
- Kept building after the semester ended, adding CSV roster management with soft-delete and audit history, per-question analytics, and a realistic seeded dataset for instructor testing.

Collegiate Hackathon — Lehigh University Fall 2025

- Built a web app converting notes and audio into flashcards and quizzes using a serverless AWS architecture (API Gateway, Lambda, Bedrock, DynamoDB) with a React + Tailwind + Vite frontend.

Cloud Computing — Course Project, Moravian University Fall 2024

- Built a serverless task-management web app using AWS Cognito, API Gateway, Lambda, and DynamoDB, deployed through AWS Amplify with a dynamic React interface.

Neural Networks — Course Project Spring 2024

- Developed a phishing-URL classifier (~95% accuracy) using Word2Vec embeddings + Ridge Classifier with tokenization and feature engineering in pandas, scikit-learn, and gensim.

Software Engineering — Course Project Spring 2024

- Built a Java MVC Monopoly game using the Factory Design Pattern, focusing on encapsulation, inheritance, and polymorphism to keep the architecture modular and testable.

Advanced Software Development — Course Project Fall 2024

- Created a Java Discord poll bot using the Discord API and Redis for real-time persistent polling with asynchronous concurrency support.

LEADERSHIP & TEACHING

Teaching Assistant & PASS Leader, CS1 (Python) · Moravian University Fall 2024 – Spring 2026

Tutor, Quantitative Resource Center (Python & Calculus I) · Moravian University 2023 – 2024

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, SQL, JavaScript, HTML/CSS

Tools & Frameworks: AWS (Lambda, API Gateway, DynamoDB, Cognito, Amplify, Bedrock), Jetpack Compose, React, Tailwind, Redis, Docker

WORK EXPERIENCE

Server / Hostess · Mama Nina's Italian Restaurant (2023–Present) · Hometown Family Restaurant (2020–2023)